

12(a)(i)	$\Delta N / \Delta T$	B1
12(a)(ii)	$\Delta N / N$	B1
12(a)(iii)	$\Delta N / (N \Delta T)$	B1
12(b)(i)	1. mass change = $5.60 \times 10^{-3} \text{ u}$	A1
	2. energy = $(\Delta)mc^2$	C1
	$= 5.6 \times 10^{-3} \times 1.66 \times 10^{-27} \times (3.0 \times 10^8)^2$ $(\text{ = } 8.36 \times 10^{-13} \text{ J})$	C1
	= 0.84 pJ	A1
12(b)(ii)	kinetic energy (of recoil) of lead (nucleus)	B1
	energy of γ -ray photon	B1